

ARYAN GOYAL

garyan18@gmail.com ◇ aryangoyal7.github.io ◇ [@arygoy](https://twitter.com/arygoy)

PUBLICATIONS AND PATENTS

Aryan Goyal. “The Hitchhiker’s Guide to Mechanistic Interpretability of Vision-Language-Action Models.” *Mechanistic Interpretability Workshop @ ICML 2026*. [\[paper\]](#)

Aryan Goyal, Ashish Mittal. “DiffusionXRay: A Diffusion and GAN-Based Approach for Enhancing Digitally Reconstructed Chest Radiographs.” *Proceedings of the Data Engineering in Medical Imaging (DEMI) Workshop @ MICCAI 2025*. [\[paper\]](#)

Aryan Goyal, Shreshtha Singh. “A Diffusion-Driven Fine-Grained Nodule Synthesis Framework for Enhanced Lung Nodule Detection from Chest Radiographs.” *Medical Imaging with Deep Learning (MIDL) 2026*. [\[paper\]](#)

Aryan Goyal. “ACADEMIA: Curriculum-Aligned Multi-Agent Orchestration for K-12 Education.” *ICLR 2026 Workshop*, Submitted.

Aryan Goyal, Qure.ai Technologies Private Limited. “A System and Method for Projecting Synthetic Nodules in Medical Imaging.” Indian Patent Application No. 202521024259 (Filed).

EDUCATION

Indian Institute of Technology Bombay

2021 - 2026

Dual Degree (B.Tech. + M.Tech.), Environmental Science and Engineering
Minor in Machine Learning and Data Science

Carnegie Mellon University

Deferred

M.S. in Computer Vision, Robotics Institute, School of Computer Science

EXPERIENCE

Independent Researcher

Jan 2026 - Present

- Research on **imitation learning** and **generalist robot policies**: isolating whether explicit dynamics prediction causally produces out-of-distribution generalization under physical parameter shift, and characterizing **stability-regime action chunking** for long-horizon manipulation.
- Secured a **\$100K Microsoft Azure compute grant** (8×A100) supporting the work; earlier threads on multi-modal action distributions in diffusion policies and mechanistic interpretability of VLAs.

Entrepreneur First | Entrepreneur-in-Residence

May 2026 - Aug 2026

- Explored the technical direction for a research lab focused on **generalist robot policies**.

Qure.AI | AI Scientist Intern

May 2024 - Dec 2025

- Worked on **diffusion** models for the **lung cancer** team: first on super-resolution of chest radiographs, then training a **DiT** model to synthesize nodules with controllable characteristics.
- Developed efficient **LoRA merging** to combine learned nodule characteristics, yielding **2 publications** and a **patent** ([end-of-internship talk](#)).

Koita Centre for Digital Health, IIT Bombay | Research Assistant

Jan 2024 - Jun 2024

- Built **automated karyotyping** pipelines for chromosome classification and segmentation with Prof. Saket Choudhary; previously fine-tuned LLMs on medical-domain data for automated medical coding.

Uptrain.AI (YC W23) | AI Research Intern

Jan 2024 - Apr 2024

- Designed novel subjective metrics for UpTrain’s **core evaluation platform for custom LLMs**, and authored the technical write-ups documenting the evaluation methodology.

UnScript.AI | Deep Learning Engineer Intern

Nov 2023 - Dec 2023

- Trained **diffusion-based text-to-speech** models for Hindi, achieving speaker voice cloning from minimal audio.

Web and Coding Club, IIT Bombay | Convener

Jul 2022 - Apr 2023

- Organised talks and hackathons, ran open-source projects, and wrote tutorials for students new to programming.

ONGOING RESEARCH PROJECTS

Measuring the stability assumption behind action chunking

Action chunking is credited with escaping compounding error. It is based on the assumption that the plant exists in an open-loop stability regime, so that error is compressive throughout the episode length. We check whether this assumption breaks by designing an algorithm that measures the stability regime, on both the open-loop and closed-loop axes. We label across the **robomimic**, **MimicGen**, **LIBERO** and **RoboCasa** datasets. We then train a predictor that estimates the open-loop stability regime for the next few timesteps, given the past.

When is implicit structure enough for dynamics modelling?

Does predicting dynamics explicitly help a policy generalize, or is the implicit structure it already learns enough? Existing work only tests this under visual distribution shift, where appearance changes but the physics do not. We test it under physical shift instead, sweeping mass, friction and centre of mass while appearance is held fixed. We compare two arms that differ in exactly one loss term, so any gain comes from the prediction objective alone. We also look at which kinds of task benefit: long-horizon, contact-rich, or failure recovery.